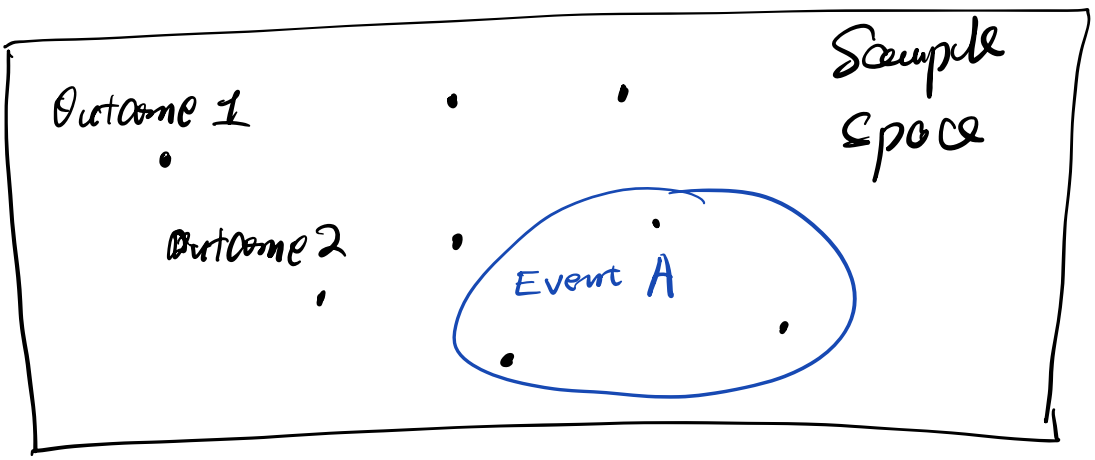
Experiment



Probability $P(A)$

Probability:Axiomatic approach

Rather than defining probability through repeated experiments, the axiomatic approach treats probability as a mathematical function satisfying certain fundamental properties.

Axiomatic Definition of Probability

Let S be the sample space. A *probability measure* P assigns a probability $P(A)$ to each event $A \subset S$ and satisfies the following axioms:

1. **Nonnegativity:**
$$P(A) \geq 0 \qquad \text{for every event } A.$$
2. **Normalization:**
$$P(S) = 1.$$
3. **Countable Additivity:**
If the sequence of events $A_1, A_2, \ldots$ are (pairwise) mutually exclusive, i.e.,
$$A_i \cap A_j = \emptyset, \qquad \forall\, i \neq j,$$
then
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots$

These three axioms are the foundation of probability theory. Many of the probability rules we will use later are consequences of these three axioms. In fact, we will build the entire theory of probability from these three axioms!

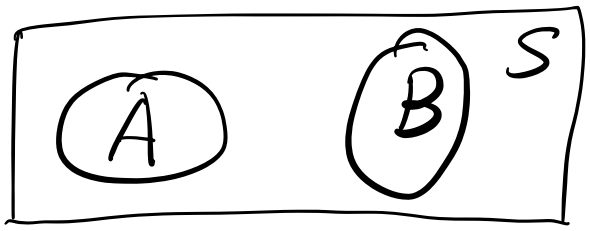
Three Axioms of Probability

For any event $A \subset S$:

$$\boxed{1.\ P(A) \geq 0} \qquad \boxed{2.\ P(S) = 1}$$

If $A_1, A_2, \ldots$ are pairwise mutually exclusive, then

$$\boxed{3.\ P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)}$$



Some Properties of Probability

- **Probability of mutually exclusive events**

  Suppose A and B are mutually exclusive:

  $$A \cap B = \emptyset.$$

  By the additivity axiom,

  $$P(A \cup B) = P(A) + P(B).$$

- **Basic consequences of the axioms**

Three Axioms of Probability
For any event $A \subset S$:

1. $P(A) \geq 0$
2. $P(S) = 1$

If $A_1, A_2, \ldots$ are pairwise mutually exclusive, then

3. $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$

$$P(\emptyset) = 0,$$

$S \cap \emptyset = \emptyset$

$\therefore$ S and $\emptyset$ are disjoint.

$P(S \cup \emptyset) = P(S) + P(\emptyset)$

$P(S) = P(S) + P(\emptyset)$

$\therefore$ $P(\emptyset) = 0.$

$$0 \leq P(A) \leq 1,$$

$A \subset S$ is any event.



we have $A \cup A^c = S$

and $A \cap A^c = \emptyset$.

$\therefore$ A and A^c is mutally exclusive.

Then $P(A \cup A^c) = P(A) + P(A^c)$

$P(S) = P(A) + P(A^c)$

$1 \geq 0$

$\therefore$ $P(A) \leq 1.$

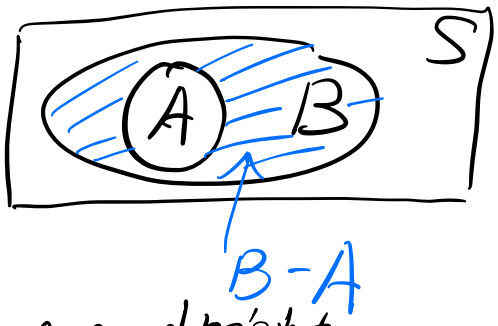
$$P(A^c) = 1 - P(A).$$

From the above discussion, we have

$P(S) = P(A) + P(A^c)$

$1 = P(A) + P(A^c)$

$\therefore$ $P(A^c) = 1 - P(A)$



- **Monotonicity**

  If $A \subset B$, then

$$P(A) \leq P(B).$$

$B = A \cup (B-A)$ and A and $B-A$ are disjoint.

$P(B) = P(A \cup (B-A)) = P(A) + P(B-A)$

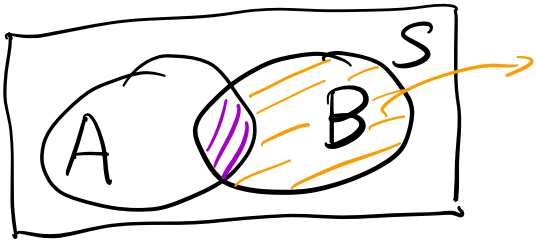
$P(B-A) \geq 0$

$\therefore P(B) \geq P(A)$

- **General addition rule**

  Suppose A and B are two events that are **not** **necessarily mutually exclusive**, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



$A \cup B = A \cup (B - A)$ where A & $B-A$ is disjoint

$P(A \cup B) = P(A \cup (B-A))$

Axiom 3 $= P(A) + P(B-A)$

$B = (B-A) \cup (A \cap B)$ where $B-A$ & $A \cap B$ are disjoint.

$P(B) = P((B-A) \cup (A \cap B))$

$= P(B-A) + P(A \cap B)$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Sometimes we write $P(A \cap B)$ as $P(AB)$.

- **Equally likely outcomes**

  In an experiment with the sample space $S = \{s_1, s_2, \cdots, s_n\}$, we say that the n outcomes are **equally likely**, if no one outcome is more likely than any other. A fair die is a familiar example of an experiment with equally likely outcomes.

  Example. Roll a fair six-sided die once. The sample space is

  $$S = \{1, 2, 3, 4, 5, 6\}.$$

  Since the six outcomes are equally likely,

  $$P(\{1\}) = P(\{2\}) = \cdots = P(\{6\}) = \frac{1}{6}.$$

  More generally, for an experiment with sample space $S = \{s_1, s_2, \cdots, s_n\}$ in which each outcome s_i is equally likely, then

  $$P(\{s_i\}) = \frac{1}{n}, \quad 1 \le i \le n.$$

$\{s_i\}$ $i = 1, 2, \cdots, n$.

$\{s_1\}, \{s_2\}, \cdots, \{s_n\}$ are pairwise disjoint, that is,

$\{s_i\} \cap \{s_j\} = \emptyset$, $\forall\, i \neq j$.

$P(\{s_1\} \cup \{s_2\} \cup \cdots \cup \{s_n\})$

$= P(\{s_1\}) + P(\{s_2\}) + \cdots + P(\{s_n\})$

$= 1$

Since $P(\{s_i\}) = P(\{s_j\})$ (equally likely events)

$\therefore$ $P(\{s_i\}) = 1/n$.

Conditional Probability

Motivating Example: Conditional Probability

Suppose a fair six-sided die is rolled once. The sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Let

$$A = \{\text{the outcome is greater than } 3\} = \{4, 5, 6\}.$$

and

$$B = \{\text{the outcome is even}\} = \{2, 4, 6\}.$$

Then,

$$P(A) = P(\{4\}) + P(\{5\}) + P(\{6\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

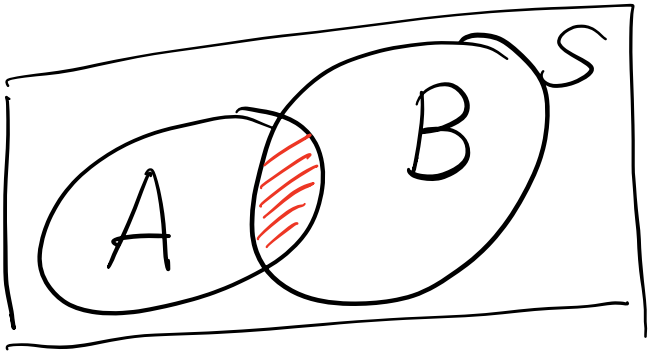
$$P(B) = 1/2$$

Now suppose we are told that the outcome is even, that is, the event B has occurred. Then, what is the probability of the event A given the fact that the event B has occurred?

Key idea: Once we know that B has occurred, the sample space is effectively reduced from S to B.

$$P(A|B) = \frac{2}{3}$$

"Probability of A given B"



$A \cap B = \{4, 6\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{1/2} = \frac{2}{3}$$

Definition: Conditional Probability

The **conditional probability** of the event A given the occurrence of the event B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$